

● HARD

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

06

Q1

ADVANCED MATH

$$h(x) = -16x^2 + 100x + 10$$

The quadratic function above models the height above the ground h , in feet, of a projectile x seconds after it had been launched vertically. If $y = h(x)$ is graphed in the xy -plane, which of the following represents the real-life meaning of the positive x -intercept of the graph?

- A. The initial height of the projectile
- B. The maximum height of the projectile
- C. The time at which the projectile reaches its maximum height
- D. The time at which the projectile hits the ground

The function g is defined by $gx = x + 14t - x$, where t is a constant. In the xy -plane, the graph of $y = gx$ passes through the point $(24, 0)$. What is the value of $g(0)$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

$$fx = 4x^2 + 64x + 262$$

The function g is defined by $gx = fx + 5$. For what value of x does gx reach its minimum?

- A. -13
- B. -8
- C. -5
- D. -3

Q4

ADVANCED MATH

The function f gives the product of a number, x , and a number that is 91 more than x . Which equation defines f ?

- A. $fx = x^2 + x + 91$
- B. $fx = x^2 + 91$
- C. $fx = x^2 + 91x$
- D. $fx = x^2 + 91x + 91$

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. $(0, 14)$
- B. $(0, 2)$
- C. $(0, 22)$
- D. $(0, -8)$

For the function f , $f0 = 86$, and for each increase in x by 1, the value of fx decreases by 80%. What is the value of $f2$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

ADVANCED MATH

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of 10 feet above the ground and reaches its maximum height of 1,034 feet above the ground 8 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 10 seconds after being launched?

- A. 234
- B. 778
- C. 970
- D. 1,014

Q8

ADVANCED MATH

$$f(x) = x^2 - 44x - 46$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

- A. 46
- B. 45
- C. 44
- D. -1

Q9

ADVANCED MATH

$$h(x) = 2(x - 4)^2 - 32$$

The quadratic function h is defined as shown. In the xy -plane, the graph of $y = h(x)$ intersects the x -axis at the points $(0, 0)$ and $(t, 0)$, where t is a constant. What is the value of t ?

- A. 1
- B. 2
- C. 4
- D. 8

Q10

ADVANCED MATH

In the xy -plane, a parabola has vertex $(9, -14)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

- A. -23
- B. -19
- C. -14
- D. -12